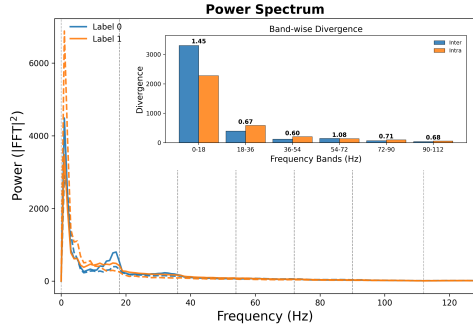


$$\mathcal{P} = \{(i, j) \mid y_i = y_j, s_i \neq s_j\}$$

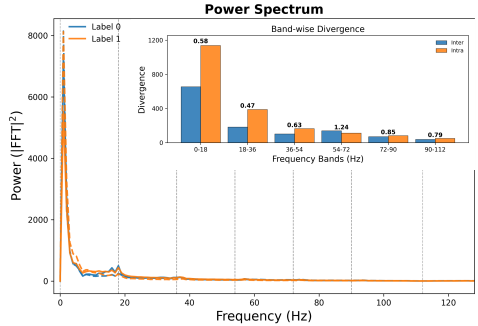
$$\mathcal{L}_{\text{band-align}} = \sum_{k=1}^K \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left\| z_i^{(k)} - z_j^{(k)} \right\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{band}} \mathcal{L}_{\text{band-align}}$$

$z_i^{(k)}$ denotes the representation of sample i in the k -th band, and the loss is computed over all label-consistent but subject-discrepant pairs within each mini-batch($\mathcal{P}$), involving all eligible samples.



(a) w/o $\mathcal{L}_{\text{band-align}}$



(b) w/ $\mathcal{L}_{\text{band-align}}$

Regarding the explicit structural alignment loss, our additional experiments suggest that it tends to over-constrain the representation. In cross-subject BTS, good alignment should reduce intra-class subject drift while preserving inter-class discriminative structure, rather than simply minimizing cross-subject discrepancy. However, a hard alignment loss can over-pull same-label samples toward a shared template, which suppresses not only nuisance variation but also informative class-related variability. This is consistent with our motivation for SCLN, where we explicitly note that overly strong spectral alignment may reduce meaningful intra-class variability by pulling same-label samples toward a common template, and with our FBD formulation, where improvement requires contraction of intra-class drift without collapsing inter-class structure.